

SHAIK BALAJI MAHAMMAD RAFI

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EXPERIENCE

Kerala Ayurveda Ltd.

Jan 2026 – Present

Agentic AI Intern

Bengaluru, India

- Architected and deployed a full-stack Next.js wellness platform featuring dynamic yoga sequence personalization, embedding multi-tier safety-first validation guardrails tailored to complex health profiles and user experience levels.
- Spearheaded a hybrid agentic RAG pipeline over 35+ traditional Ayurvedic texts, utilizing custom OCR data extraction, advanced chunking, vector indexing, and precise tool-calling architectures for clinical decision support.
- Boosted Shopify storefront productivity by autonomously integrating AI-driven code generation, optimizing high-conversion PDP and blog modules to accelerate web deployment cycles.
- Conducted data-backed competitive intelligence and product positioning analyses, mapping digital assets to drive market ranking strategies and enterprise growth.

IIT Ropar

May 2025 – July 2025

Summer Intern

Punjab, India (Remote)

- Architected scalable MERN-stack frontend interfaces and low-latency state management layers for a real-time, interactive academic polling platform.
- Engineered an automated speech-to-poll pipeline combining OpenAI Whisper live audio transcription with structured LLM prompt orchestration for instant question generation.

SKILLS

Languages: Python (Expert), TypeScript, Java

AI/LLM Frameworks: LangGraph, LangChain, LlamaIndex, PyTorch/TensorFlow

Data & Vector DBs: MongoDB, Supabase, Pinecone, ChromaDB

Full-Stack: React, Next.js, Node.js, FastAPI

Infrastructure & MLOps: Docker, GCP/AWS, GitHub Actions, CI/CD

Observability & Evaluation: LangSmith, Weights & Biases

PROJECTS

Know Your Parents — Local-First Family Archive & AI Platform 🐙

April 2026

- Designed a Next.js 16/FastAPI full-stack platform featuring a hybrid GraphRAG pipeline that combines lexical search, ChromaDB vector indexing, and NetworkX graph-distance weighting.
- Integrated multimodal ingestion workflows using faster-whisper transcription and OCR, backed by grounded multi-stage agentic reasoning with strict citation guardrails.

Project Phoenix — Autonomous Self-Healing System 🐙

Dec 2025 – March 2026

- Architected a zero-touch recovery framework featuring multi-agent coordination, self-correction loops, validation gates, and automated rollbacks for autonomous system remediation.
- Built full-stack observability layers for long-running agents operating under non-deterministic environments, enabling real-time state tracking, telemetry, and root-cause diagnosis.

Project Chimera — Open-Source Space Biology Research Engine 🐙

October 2025

- Spearheaded a 5-agent NASA research platform using LangGraph state graphs (StateGraph) with real-time SSE execution streaming and dynamic OSDR/GeneLab API connectors.
- Engineered a 5-stage hybrid retrieval system combining FAISS vector search, keyword matching, Reciprocal Rank Fusion (RRF), Cross-Encoder reranking, and local Ollama LLM execution.

EDUCATION

VVIT, Guntur

April 2026

B.Tech in Computer Science and Engineering

CGPA: 8.97 / 10

CERTIFICATIONS

AI & LLMs: Prompt Engineering & Programming with OpenAI (Columbia+), Responsible & Safe AI Systems (NPTEL)

Cloud & Analytics: Google Cloud Data Analytics, Cloud Computing, Data Analytics with Python

Cross-Functional: Business Japanese & Business Manner, Japanese Language & Culture I–II, International B2B Marketing

LEADERSHIP & AWARDS

- **Grand Finalist** at **Google Cloud Agentic AI Day 2025**, selected from **57,000+ developers** at a **Guinness World Record** national event by Google Cloud, Google for Developers, FI & Hack2Skill.
- Recognized as an **NPTEL Discipline Star** and invited to **IIT Madras** for academic excellence.
- ACM Officer: Designed coding contests and supported national execution of **Spardha 2K24** for **6,500+ participants**.
- Institution's Innovation Council: Led hackathons and incubation initiatives advancing innovation and entrepreneurship.